Stop the engine in order to check the oil level. DO NOT check the oil level when the engine is running.

Park the machine on a level surface.

1. Open the engine compartment.

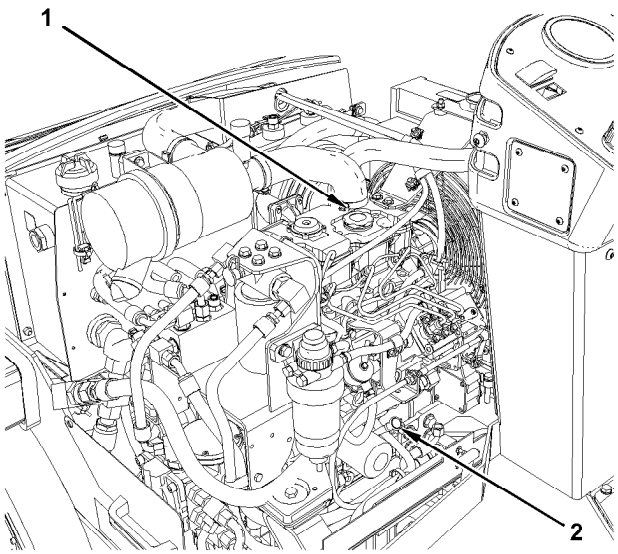


Illustration 91 g01033885

2. Remove dipstick (2). Wipe the dipstick with a clean cloth. Insert the dipstick. Remove the dipstick and note the oil level. Insert the dipstick.

Note: Refer to the Operation and Maintenance Manual, “Lubricant Viscosities” and Operation and Maintenance Manual, “Capacities (Refill)” for the correct amount of oil that is used when the oil is changed. The correct amount of oil determines the correct level of the oil in the FULL range on the dipstick.

NOTICE
Do not overfill the crankcase. The oil level must not reach the top of the **FULL** range mark or above the **FULL** range mark.

3. Maintain the oil level on the dipstick between the MINIMUM mark and the MAXIMUM mark . Add oil if the oil level is too low. If the oil level is correct, close the engine compartment.

Note: Operating your engine with the oil level above the FULL mark in the FULL Range could cause the crankshaft to dip into the oil. This could result in excessively high operating temperatures. The high operating temperatures could result in reduced lubricating characteristics of the oil. This could cause damage to the bearings and loss of engine power.

Add The Engine Oil

1. Open the engine compartment.

2. Remove oil filler cap (1).

3. Add the oil.

4. Clean oil filler cap (1). Install oil filler cap (1).

5. Close the engine compartment.

i01175145

Engine Oil Sample - Obtain

SMCS Code: 1000-008

⚠ WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact the skin.

Obtain the Sample and the Analysis

In addition to a good preventive maintenance program, Caterpillar recommends using S·O·S oil analysis at regular scheduled intervals in order to monitor the condition of the engine and the maintenance requirements of the engine.

Each oil sample should be taken when the oil is warm and when the oil is well mixed. The sample should be taken at this time in order to ensure that the sample is representative of the oil in the crankcase.

Obtain the S·O·S Sample

Use the following method in order to obtain an S·O·S sample:

- Use a 1U-5718 Vacuum Pump or use an equivalent pump that is inserted into the sump.

To avoid contamination of the oil samples, the tools and the supplies that are used for obtaining oil samples must be clean.

Consult your Caterpillar dealer for complete information and assistance in establishing an S·O·S program for your engine.

If you fill the engine too fast with oil, the oil may saturate the engine breather. If the breather is saturated with oil, oil will blow out of the breather hose until the breather is free of oil. Add the engine oil at a rate of 2 L/min (0.5283 US gpm). This will help prevent saturating the breather with oil.

i01977942

Engine Oil and Filter - Change

SMCS Code: 1318-510

Run the engine in order to warm up the oil. Stop the engine before you drain the oil. When the oil is warm the waste particles are suspended in the oil. The waste particles will be removed when the oil is drained.

As the oil cools, the waste particles settle to the bottom of the oil pan. The waste particles will not be removed if the oil is too cool.

The waste particles can recirculate through the engine lubrication system if the recommended procedure is not followed.

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

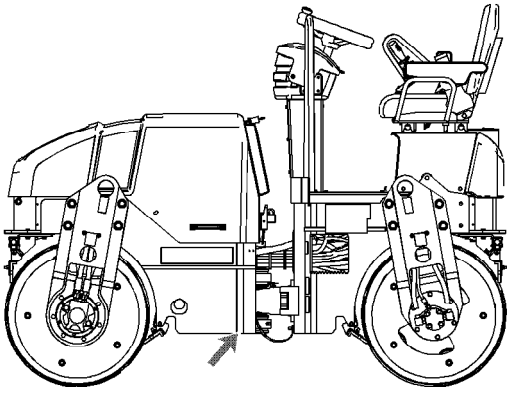


Illustration 92 g01033955

2. Place a suitable container under the drain hose. The drain hose is located below the front frame. Open the drain valve. Refer to the Operation and Maintenance Manual, “General Hazard Information” for information that pertains to containing fluid spillage.

3. Allow the oil to completely drain.

4. Close the drain valve.

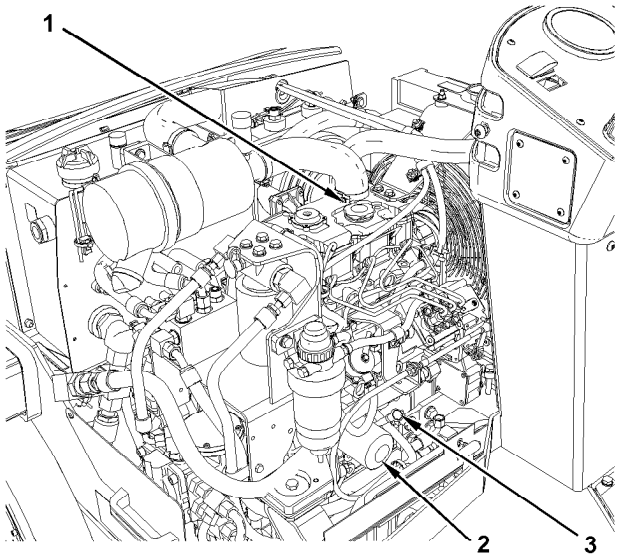


Illustration 93 g01033957

5. Remove filter element (2).

Note: Discard the used filter element according to local regulations.

6. Clean the filter housing base. All of the old filter seal must be removed from the filter housing base.

7. Apply a thin coat of engine oil to the seal of the new filter element.

8. Install the new filter by hand. When the gasket contacts the filter base, tighten the filter element for an additional 3/4 turn. This will tighten the filter sufficiently.

Every new oil filter has rotation index marks that are spaced at 90 degree increments. Use the rotation index marks as a guide for tightening the oil filter.

9. Remove oil filler cap (1). Fill the crankcase with new oil. See Operation and Maintenance Manual, “Lubricant Viscosities” and Operation and Maintenance Manual, “Capacities (Refill)” for further information. Clean the oil filler cap and install the oil filler cap.

10. Before you start the engine, check the oil level on dipstick (3). The oil level must be within the FULL RANGE on the dipstick.

11. Start the engine. Run the engine for two minutes. Inspect the machine for leaks. Stop the machine.

12. Wait for ten minutes in order to allow the oil to drain back into the crankcase. Check the oil level. Maintain the oil level within the FULL RANGE on the dipstick.

13. Close the engine compartment.

i01633611

Engine Valve Lash - Check

SMCS Code: 1105-535

This maintenance is recommended by Caterpillar as part of a lubrication and preventive maintenance schedule in order to help provide maximum engine life.

NOTICE
Only qualified service personnel should perform this maintenance. Refer to the Service Manual or your Caterpillar dealer for the complete valve lash adjustment procedure.

Operation of Caterpillar engines with improper valve adjustments can reduce engine efficiency. This reduced efficiency could result in excessive fuel usage and/or shortened engine component life.

i02166043

Fuel System - Prime

SMCS Code: 1250-548

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, “Caterpillar Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

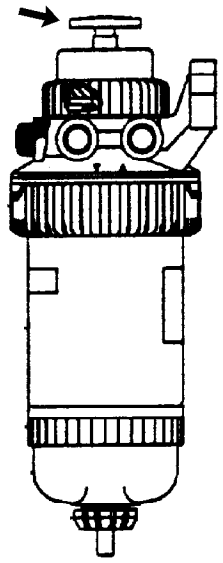


Illustration 94 g01098216

1. Operate the fuel priming pump plunger in order to fill the new filter element with fuel. Continue to pump until a resistance is felt. This resistance will indicate that the filter element is full of fuel.

2. Start the engine. If the engine will not start, further priming is necessary. If the engine starts but the engine continues to misfire, further priming is necessary. If the engine starts but the engine continues to emit smoke, further priming is necessary.

3. If the engine starts but the engine runs rough, continue to run the engine at low idle. Continue to run the engine at low idle until the engine runs smoothly.

i02960208

Fuel System Primary Filter (Water Separator) Element - Replace

SMCS Code: 1260-510-FQ; 1263-510-FQ

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

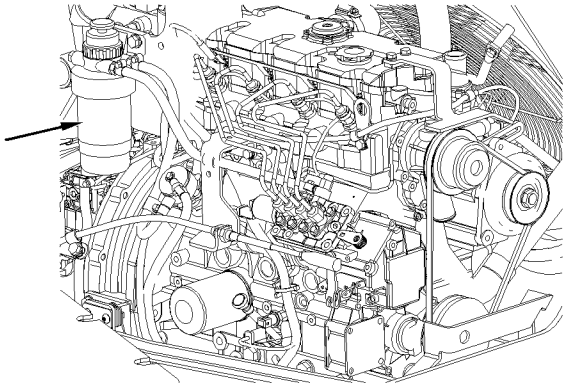


Illustration 95 g01224649

Location of the fuel system water separator

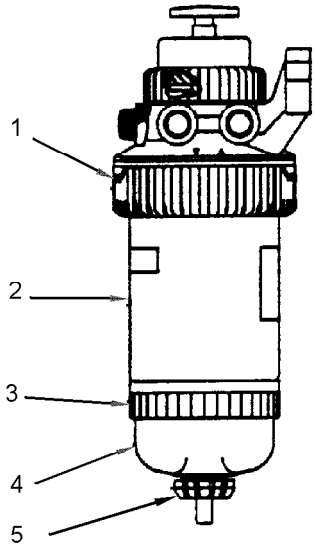


Illustration 96 g01034054

2. Turn valve (5) in a counterclockwise direction in order to drain the fuel. Drain the fuel into a suitable container.

Note: Dispose of the waste in a proper manner.

3. Close valve (5).

4. Hold bowl (4) while you loosen the collar (3). Remove the bowl (4) and the collar (3).

5. Loosen the collar (1). Remove the element (2). Discard the element in a proper manner.

6. Clean the parts. Inspect the parts for damage. Replace the damaged parts or replace the worn parts.

7. Clean the filter mounting base. All of the old seal must be removed.

8. Apply a thin coat of clean diesel fuel to the seal on the new filter. Install a new fuel filter hand tight until the seal of the filter contacts the base.

9. Tighten the filter according to the instructions that are printed on the filter. Tighten the collar (1).

10. Install the bowl (4) and tighten the collar (3).

11. Start the engine and check the fuel system for leaks.

12. Close the engine compartment.

i02457222

Fuel System Water Separator - Drain

SMCS Code: 1263-543; 1263

1. Lift the engine compartment. refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

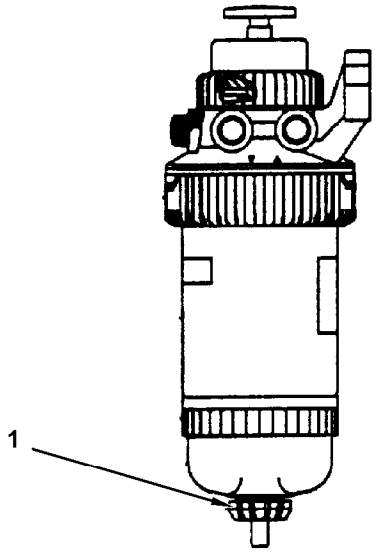


Illustration 97 g01034098

Note: Attach a rubber drain tube to valve (1) in order to avoid spilling water on the muffler.

2. Turn valve (1) counterclockwise in order to open the valve.

3. Drain the water and the sediment into a suitable container.

Note: Always dispose of drained fluids in accordance with local regulations.

4. Close valve (1).

5. Remove the rubber drain tube.

6. Close the engine compartment.

i01978018

Fuel Tank Cap and Strainer - Clean

SMCS Code: 1273-070-Z2; 1273-070-STR

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

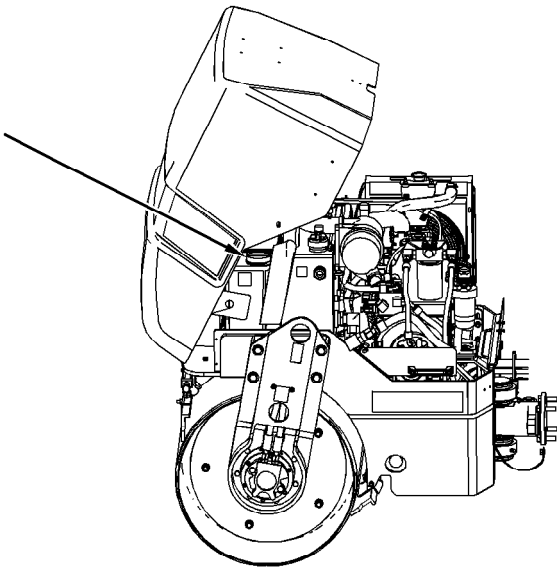


Illustration 98 g01034106

2. Remove fuel tank cap.

3. Remove the filler screen.

4. Wash the filler screen in clean, nonflammable solvent. Dry the filler screen with pressure air.

5. Inspect the cap and the filler screen. Replace the cap if the cap is damaged. Replace the filler screen if the filler screen is damaged.

6. Install the filler screen.

7. Apply a thin film of fuel to the gasket of the fuel tank cap.

8. Install fuel tank cap.

9. Close the engine compartment.

i01978021

Fuel Tank Water and Sediment - Drain

SMCS Code: 1273-543-M&S

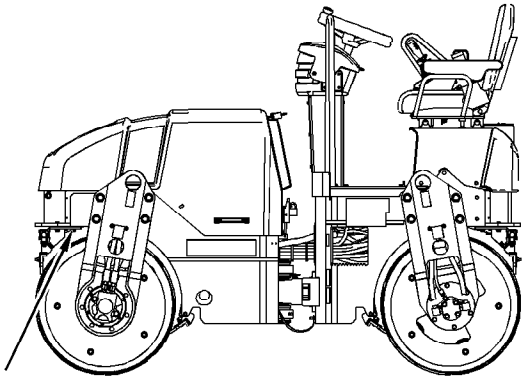


Illustration 99 g01034194

1. The drain plug is located under the left front side of the machine.

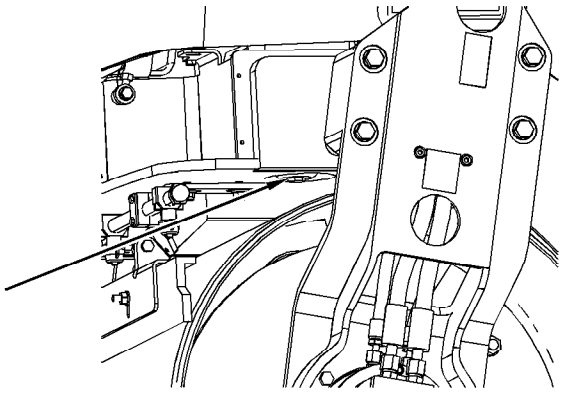


Illustration 100 g01034210

2. Remove the drain plug. Allow the water and sediment to drain into a suitable container.

3. Install the drain plug.

Note: Dispose of all fluids according to local regulations.

i01978023

Fuses - Replace

SMCS Code: 1417-510

Fuse – The fuses protect the electrical system from damage that is caused by overloaded circuits. Change the fuse if the element separates. If the element of the new fuse separates, check the circuit. Repair the problem before you operate the machine.

NOTICE
Replace fuses with the same type and size. Improper use of fuses could result in electrical damage. Frequent replacement of fuses may indicate another type of electrical problem. Contact your Caterpillar dealer.

The compartment for the fuses is located on the right side of the machine. There are four screws that hold the cover on the compartment.

In order to access the compartment for the fuses, remove the four screws. Remove the cover.

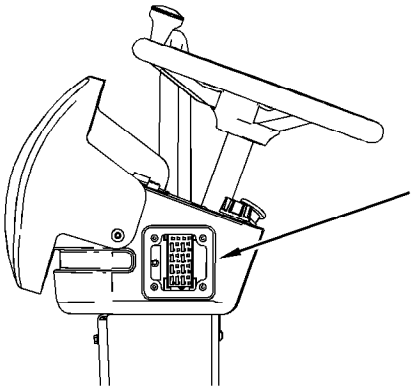


Illustration 101 g01034791

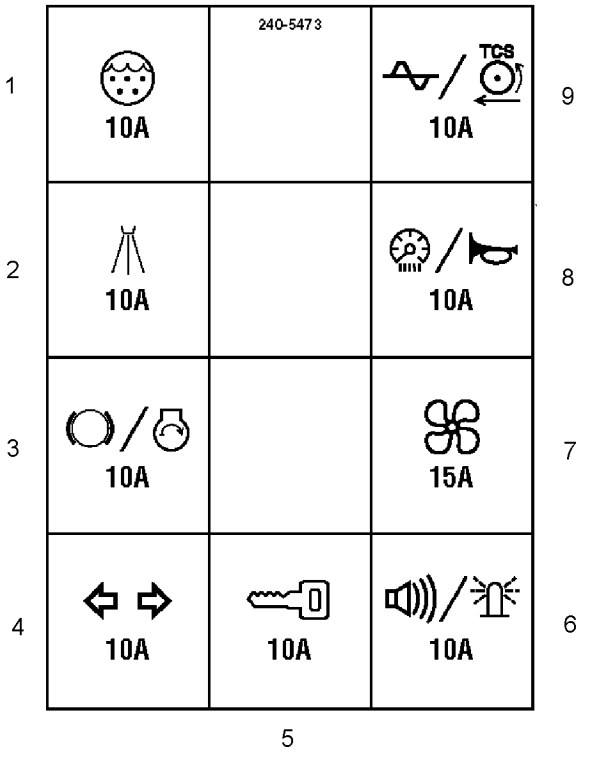


Illustration 102 g01034810

Emulsion Spray System (1) (If Equipped) – 10 amp

Water Spray System (2) – 10 amp

Brake and Neutralizer (3) – 10 amp

Flashers (4) – 10 amp

Keyswitch (5) – 10 amp

Backup Alarm and Beacon Light (6) (If Equipped) – 10 amp

Oil Cooler Fan (7) – 15 amp

Gauges and Horn (8) – 10 amp

Vibratory System and Flow Divider (9) (If Equipped) – 10 amp

i01978026

Hydraulic Oil Cooler - Clean

SMCS Code: 1374-070

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

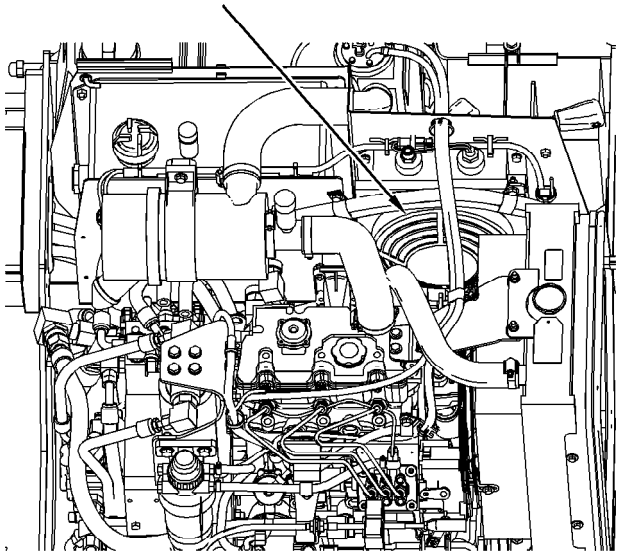


Illustration 103 g01034826

2. Inspect the condition of the oil cooler. Clean the oil cooler. Compressed air, high pressure water or steam can be used. Remove debris from the oil cooler.

3. Close the engine compartment.

i01978027

Hydraulic System Oil - Change

SMCS Code: 5050-044; 5095-044

NOTICE
Take extreme care to insure the cleanliness of the hydraulic oil. Keep the hydraulic oil clean in order to extend the component life and assure the maximum performance.

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

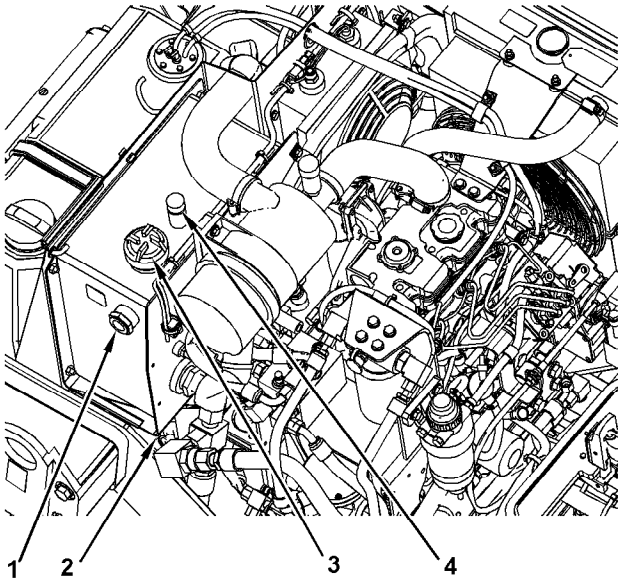


Illustration 104 g01034853

2. Remove hydraulic tank filler cap (3).

3. Remove the screen from the filler tube of the hydraulic tank.

4. Wash hydraulic tank filler cap (3) in clean, nonflammable solvent. Wash the screen in clean, nonflammable solvent.

5. Check the vent for the hydraulic tank (4). Clean the vent for the hydraulic tank with clean, nonflammable solvent.

Note: The drain for the hydraulic tank oil (2) is located on the left side of the engine compartment. A drain hose is connected to the drain on the left side of the frame.

6. Open the drain for the hydraulic tank (2) in order to drain the oil. Drain the oil in a suitable container.

7. Remove the suction strainer inside the hydraulic tank. Install a new suction strainer in the hydraulic tank.

8. Close the drain for the hydraulic tank (2).

9. Install the screen in the hydraulic tank.

10. Refill the hydraulic tank with clean, filtered hydraulic oil. Refer to the Operation and Maintenance Manual, “Capacities (Refill)” and Operation and Maintenance Manual, “Lubricant Viscosities”.

11. Park the machine on level ground. Check the hydraulic oil level. The oil level should be visible in the sight gauge (1).

Refer to Operation and Maintenance Manual, “Hydraulic System Oil Level - Check”.

12. Install hydraulic tank filler cap (3).

13. Close the engine compartment.

i01999970

Hydraulic System Oil Filter - Replace

SMCS Code: 5068-510

⚠ WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, NENG2500, Caterpillar Tools and Shop Products Guide for tools and supplies suitable to collect and contain fluids on Caterpillar products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Take extreme care to insure the cleanliness of the hydraulic oil. Keep the hydraulic oil clean in order to extend the component life and assure the maximum performance.

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

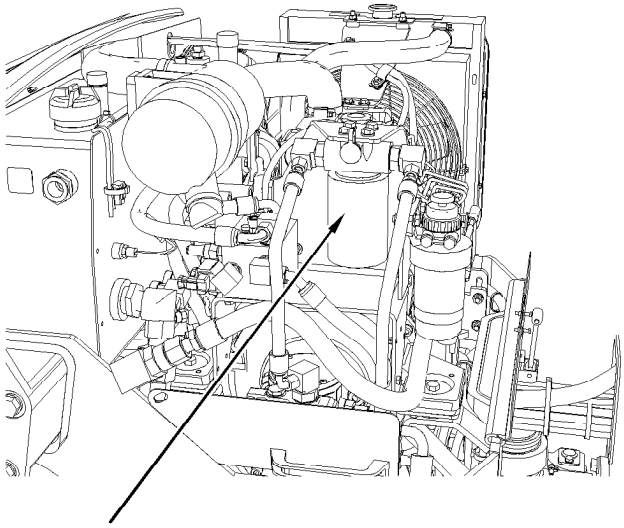


Illustration 105 g01034862

2. Clean the hydraulic oil filter location.

3. Remove the filter with a strap type wrench.

4. Clean the filter housing base. Remove any existing gasket material.

5. Apply a light coat of hydraulic oil to the gasket on the new filter.

6. Use your hand to install the new filter. When the seal contacts the base, tighten the filter element for an additional three quarters of a turn.

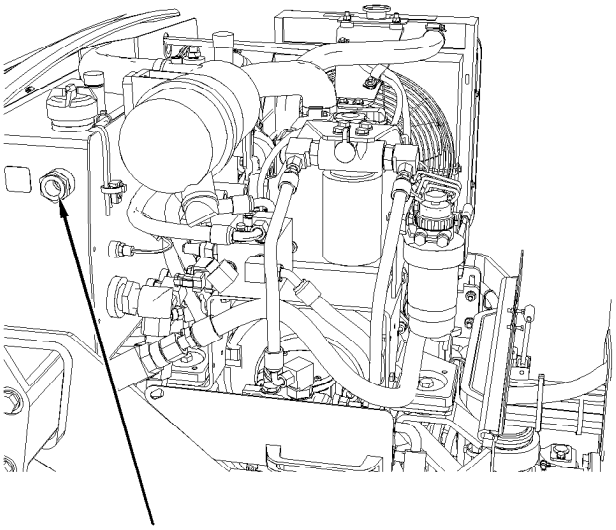


Illustration 106 g01034864

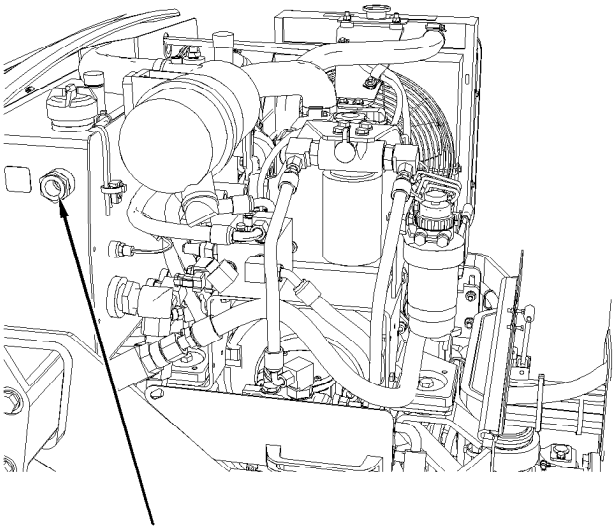


Illustration 107 g01034864

7. With the machine on level ground, check the hydraulic oil level in the sight gauge. The oil level should be visible in the sight gauge. Refer to the Operation and Maintenance Manual, “Hydraulic System Oil Level - Check”.

8. Add required oil.

9. Close the engine compartment.

i01978034

Hydraulic System Oil Level - Check

SMCS Code: 5056-535-FLV; 5095-535-FLV

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

2. Observe the level of the hydraulic oil in the sight gauge when the oil is warm. Maintain the oil level to the mark on the sight gauge.

3. If necessary, add oil.

Refer to the Operation and Maintenance Manual, “Capacities (Refill)” and Operation and Maintenance Manual, “Lubricant Viscosities”.

4. Close the engine compartment.

i01978039

Hydraulic System Oil Sample - Obtain

SMCS Code: 5050-008; 5056-008; 5095-008

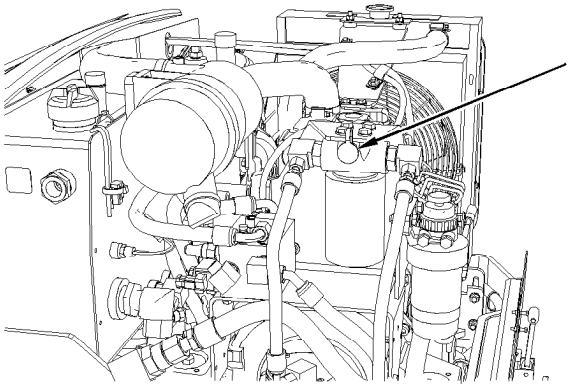


Illustration 108 g01034893

Use the sampling valve for the hydraulic oil in order to obtain the oil sample. Refer to Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations” for further information on obtaining an oil sample.

i01978041

Hydraulic Tank Breather - Replace

SMCS Code: 5050-510-BRE; 5056-510-BRE; 5118-510

Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

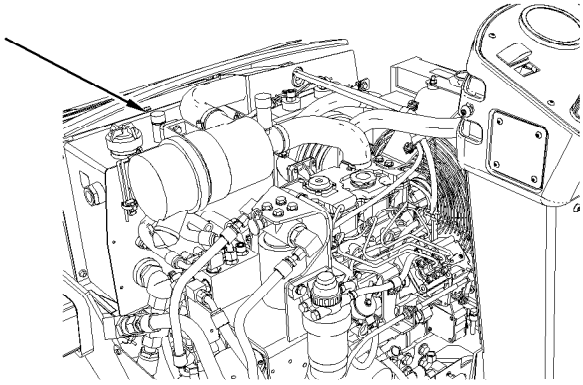


Illustration 109 g01034904

The vent is located on the top of the hydraulic tank.

Remove the vent from the hydraulic tank.

Install the new vent.

Close the engine compartment.

i02072906

Hydraulic Tank Strainer - Clean

SMCS Code: 5056-070-STR

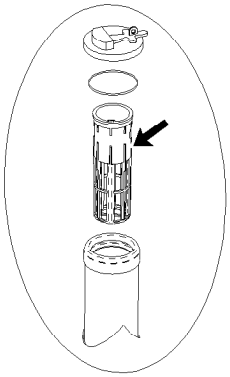


Illustration 110 g01059072

1. Remove the filler cap for the hydraulic tank.
2. Pull the strainer for the hydraulic tank out of the filler tube.
3. Clean the strainer in clean, nonflammable solvent. Dry the strainer with compressed air.
4. Install the strainer and the filler cap.

i01978047

Indicators and Gauges - Test

SMCS Code: 7450-081

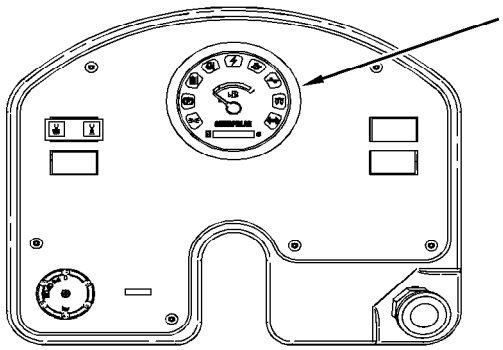


Illustration 111 g01035108

1. Look for broken lenses on the gauges, broken indicator lights, broken switches, and other broken components in the operator station.

2. Start the engine.

3. Look for inoperative gauges.

4. Turn on all machine lights. Check for proper operation.

5. Stop the engine.

Note: When the engine is stopped and the engine start switch key is turned to the ON position, all of the indicator lights should illuminate. If the indicator lights do not illuminate, replace the lights.

6. Make any repairs that are required before operating the machine.

i02024019

Isolation Mounts - Inspect

SMCS Code: 5654-040

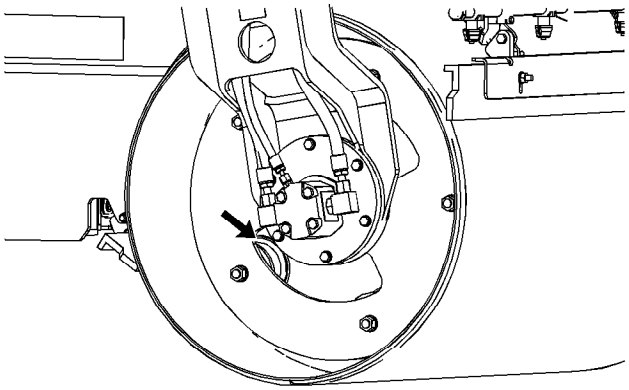


Illustration 112 g01045159

Inspect the isolation mounts for damage, cracking, or splitting. If an isolation mount is damaged, replace the mount. If two or more of the isolation mounts are damaged, replace all of the isolation mounts. Refer to the Disassembly and Assembly Manual for your machine for further information on removing and installing the isolation mounts.

i02038351

Neutral Start Switch - Test
(If Equipped)

SMCS Code: 1424-025; 1424-081; 1424-535

⚠ WARNING

The machine may lurch forward if the neutral start switch is out of adjustment. Be sure the area is clear of all personnel and equipment before performing this test.

1. Depress the parking brake knob in order to engage the parking brake.

2. Move the propel lever to the FORWARD position. Hold the engine start switch in the START position. Slowly move the propel lever toward the NEUTRAL position.

3. If the engine starts before you move the propel lever to the NEUTRAL position, the neutral start switch requires adjustment. Do not operate the machine until the repairs have been made. Refer to the service manual for your machine for the procedure on neutral start switch adjustment.

i02106227

Oil Filter - Inspect

SMCS Code: 1308-507; 3004-507; 3067-507; 5068-507

Inspect a Used Filter for Debris

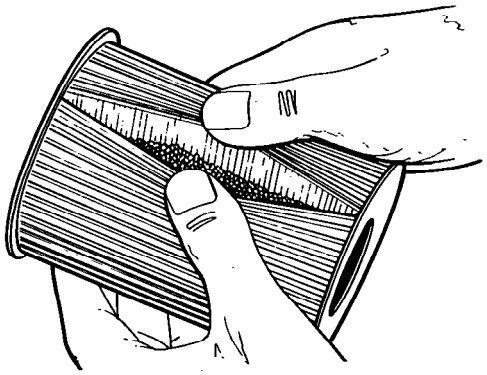


Illustration 113 g00100013

The element is shown with debris.

Use a filter cutter to cut the filter element open. Spread apart the pleats and inspect the element for metal and for other debris. An excessive amount of debris in the filter element can indicate a possible failure.

If metals are found in the filter element, a magnet can be used to differentiate between ferrous metals and nonferrous metals.

Ferrous metals can indicate wear on steel parts and on cast iron parts.

Nonferrous metals can indicate wear on the aluminum parts of the engine such as main bearings, rod bearings, or turbocharger bearings.

Small amounts of debris may be found in the filter element. This could be caused by friction and by normal wear. Consult your Caterpillar dealer in order to arrange for further analysis if an excessive amount of debris is found.

Using an oil filter element that is not recommended by Caterpillar can result in severe engine damage to engine bearings, to the crankshaft, and to other parts. This can result in larger particles in unfiltered oil. The particles could enter the lubricating system and the particles could cause damage.

i02018610

Propel Control Tension - Adjust

SMCS Code: 3209-025

1. Remove the clips that hold the rubber boot for the propel control.

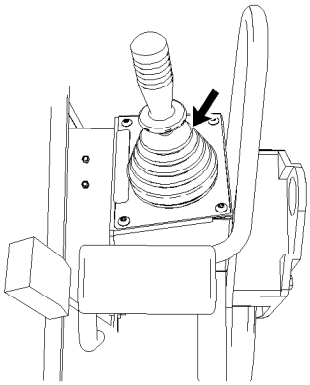


Illustration 114 g01043056

2. Remove the rubber boot.

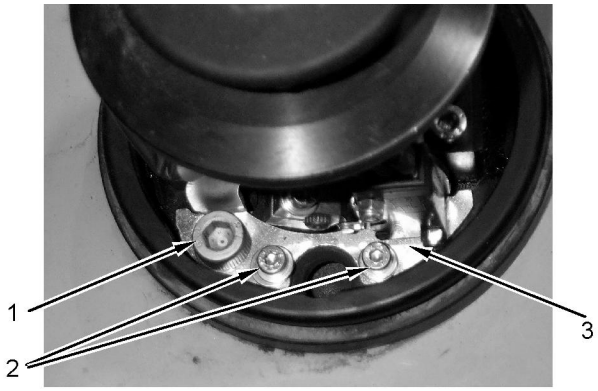


Illustration 115 g01043098

3. Remove bolt (1) and screws (2) that hold the mounting bracket for the reverse switch (3).

4. Remove the mounting bracket for the reverse switch (3).

5. Push the propel control lever to the full FORWARD position. Hold the propel control lever in the full FORWARD position.

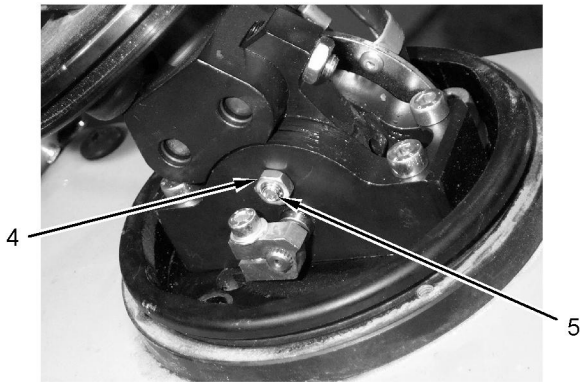


Illustration 116 g01043036

6. Loosen locknut (4).

7. Tighten adjustment screw (5) so that the propel control lever remains in position. Tighten the adjustment screw for an additional 1/4 turn.

Note: The propel control lever should remain in the full FORWARD position after you release the propel control lever.

8. Tighten locknut (4).

9. Install mounting bracket (3) for the reverse switch. Tighten bolt (1) to 6 N·m (53 lb in). Tighten screws (2) to 2.5 N·m (22 lb in).

Note: Verify that the backup alarm activates properly after the mounting bracket is installed. Do not operate the machine if the backup alarm does not activate.

10. Verify that the propel control lever will remain in place when you move the propel control lever to the full FORWARD position. Verify that the propel control lever will remain in place when you move the propel control lever to the full REVERSE position.

11. Install the rubber boot. Install the clips that hold the rubber boot in place.

i02014868

Radiator Core - Clean

SMCS Code: 1353-070-KO

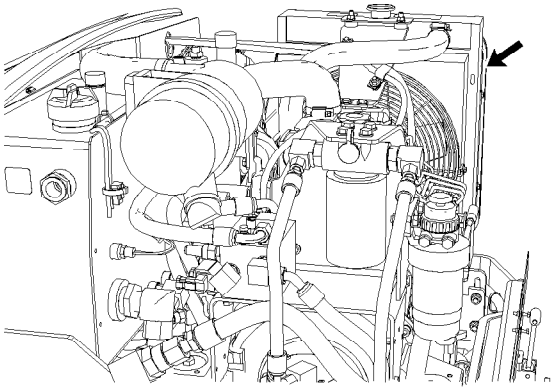


Illustration 117 g01041621

Open the engine compartment.

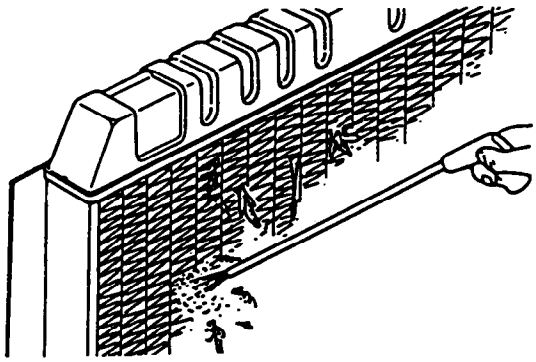


Illustration 118 g00101939

Inspect the radiator core for debris. If necessary, clean the radiator.

Compressed air is preferred, but high pressure water or steam can be used to remove dust and general debris from a radiator. Clean the radiator according to the condition of the radiator.

Note: High pressure water can bend the oil cooler and the radiator fins.

See Special Publication, SEBD0518, “Know Your Cooling System” for more information about cleaning radiator fins.

Close the engine compartment.

i02202621

Rollover Protective Structure (ROPS) - Inspect

SMCS Code: 7323-040

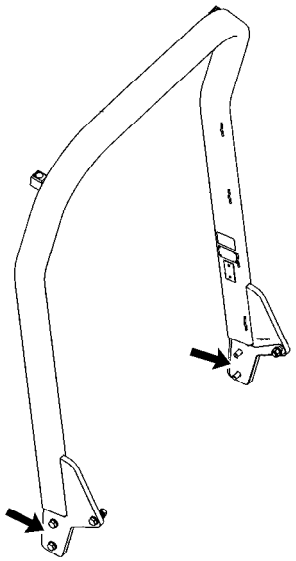


Illustration 119 g01037365
Standard ROPS

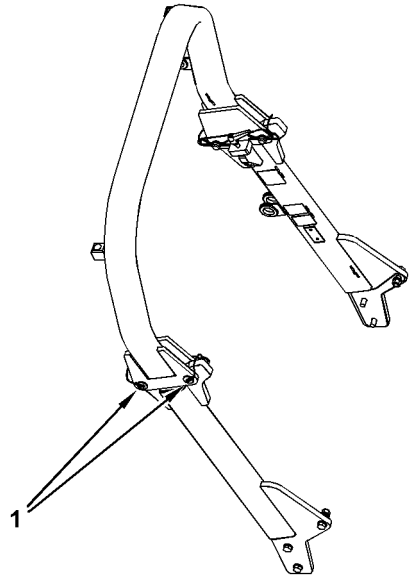


Illustration 120 g01112161
Foldable ROPS

Inspect the rollover protective structure (ROPS) for cracks. Inspect the ROPS for any loose bolts or damaged bolts. Replace the damaged bolts with original equipment parts only.

Note: If your machine is equipped with a foldable ROPS, inspect locking pins (1) for wear or damage.

Note: Apply oil to all ROPS bolt threads before you install the bolts. Failure to apply the oil to the threads can result in an improper bolt torque.

Do not straighten the ROPS or repair the ROPS by welding reinforcement plates to the ROPS.

Consult your Caterpillar dealer for the repair of the ROPS.

i04423622

Seat Belt - Inspect

SMCS Code: 7327-040

Always inspect the condition of the seat belt and the condition of the seat belt mounting hardware before you operate the machine. Replace any parts that are damaged or worn before you operate the machine.

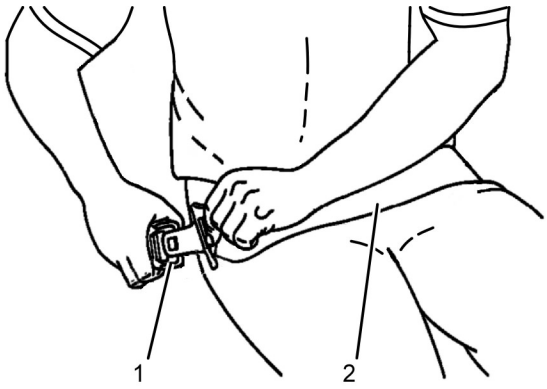


Illustration 121 g02620101
Typical example

Inspect buckle (1) for wear or for damage. If the buckle is worn or damaged, replace the seat belt.

Inspect seat belt (2) for webbing that is worn or frayed. Replace the seat belt if the webbing is worn or frayed.

Inspect all seat belt mounting hardware for wear or for damage. Replace any mounting hardware that is worn or damaged. Make sure that the mounting bolts are tight.

If your machine is equipped with a seat belt extension, also perform this inspection procedure for the seat belt extension.

Contact your Cat dealer for the replacement of the seat belt and the mounting hardware.

Note: The seat belt should be replaced within 3 years of the date of installation. A date of installation label is attached to the seat belt retractor and buckle. If the date of installation label is missing, replace belt within 3 years from the year of manufacture as indicated on belt webbing label, buckle housing, or installation tags (non-retractable belts).

i06891605

Seat Belt - Replace

SMCS Code: 7327-510

The seat belt should be replaced within 3 years of the date of installation. A date of installation label is attached to the seat belt retractor and buckle. If the date of installation label is missing, replace belt within 3 years from the year of manufacture as indicated on belt webbing label, buckle housing, or installation tags (non-retractable belts).

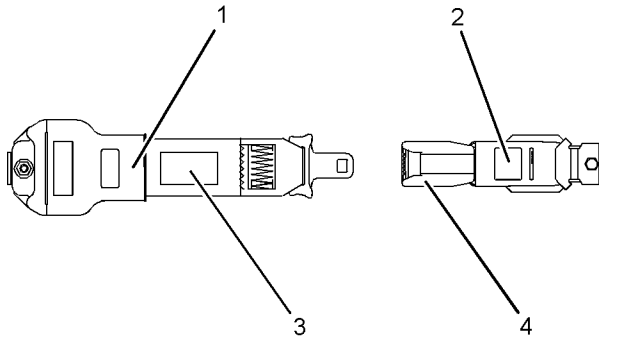


Illustration 122 g01152685

Typical Example

(1) Date of installation (retractor)
(2) Date of installation (buckle)
(3) Year of manufacture (tag) (fully extended web)
(4) Year of manufacture (underside) (buckle)

Consult your Cat dealer for the replacement of the seat belt and the mounting hardware.

Determine age of new seat belt before installing on seat. A manufacture label is on belt webbing and imprinted on belt buckle. Do not exceed install by date on label.

Complete seat belt system should be installed with new mounting hardware.

Date of installation labels should be marked and affixed to the seat belt retractor and buckle.

Note: Date of installation labels should be permanently marked by punch (retractable belt) or stamp (non-retractable belt).

If your machine is equipped with a seat belt extension, also perform this replacement procedure for the seat belt extension.

i01978077

Steering Cylinder Ends - Lubricate

SMCS Code: 4303-086-BD

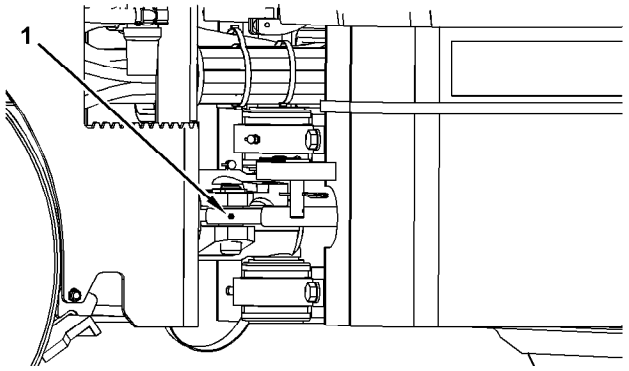


Illustration 123 g01035157

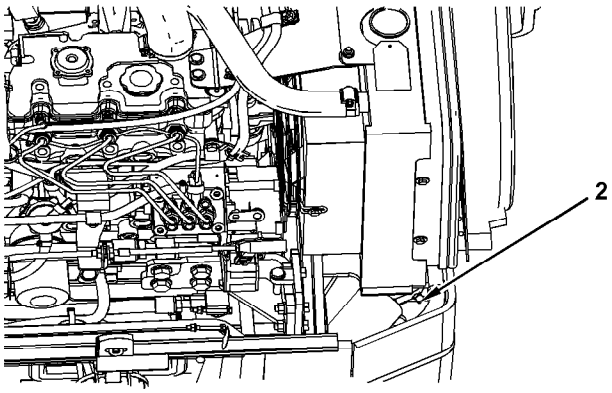


Illustration 124 g01035177

The steering cylinder is located in the pivot area. The steering cylinder is on the right side of the engine compartment. Lubricate the fitting at each end of the steering cylinder.

Note: Wipe all of the fittings before you lubricate the fittings.

i01978079

Throttle Control - Lubricate

SMCS Code: 1265-086

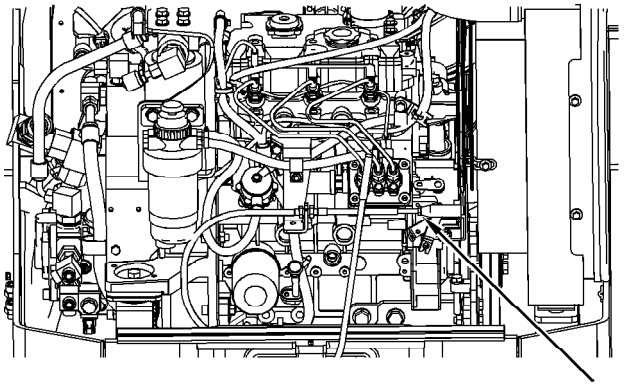


Illustration 125 g01035231

1. Open the engine compartment. Refer to the Operation and Maintenance Manual, “Access Doors and Covers”.

2. Clean the throttle control linkage with a clean rag.

3. Lubricate the throttle control linkage with engine oil.

4. Close the engine compartment.

i01978085

Tire Inflation - Check

SMCS Code: 4203-535-PX

S/N: 2251–Up

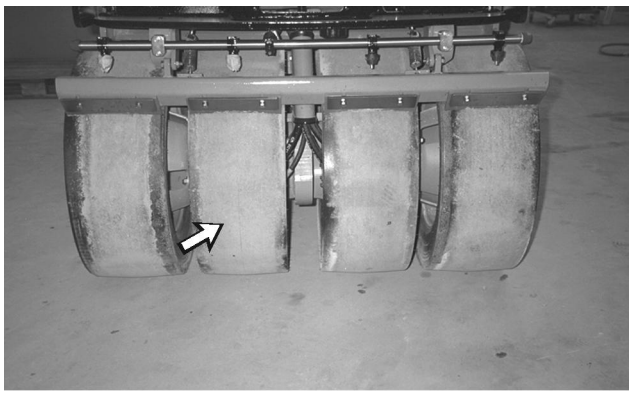


Illustration 126 g00668495

If necessary, inflate the tires.

For normal operating conditions, inflate the tires to the proper pressure:

Table 11

Tire Size	Minimum Pressure	Maximum Pressure
9.5/65-15	260 kPa (38 psi)	325 kPa (47 psi)
8.5/90-15	170 kPa (25 psi)	350 kPa (50 psi)

The tire inflation pressure is based on the weight of a machine that is ready to work and the machine is without attachments. The pressure is based on the rated payload, and in average operating conditions. Pressures for each application may vary. Consult your Caterpillar dealer for more information about proper tire inflation pressures.

i01978087

Tire Scraper - Inspect/Adjust/ Replace

SMCS Code: 6607-040; 6607-510; 6607-025

S/N: 2251–Up

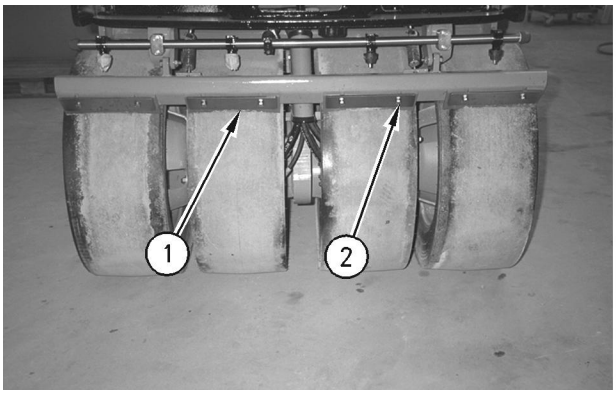


Illustration 127 g00646008

1. Clean the tire scrapers (1).

2. Adjust the tire scrapers (1).

3. Replace the tire scrapers if the tire scrapers are damaged.

4. Loosen the bolts (2) that fasten the tire scrapers.

5. Remove the damaged tire scraper (1).

6. Install the new tire scraper.

7. Tighten the bolts on the new tire scraper.

8. Adjust the new tire scraper so that the new tire scraper rides just above the tires.

i01978101

Water Spray Nozzles - Clean

SMCS Code: 6609-070

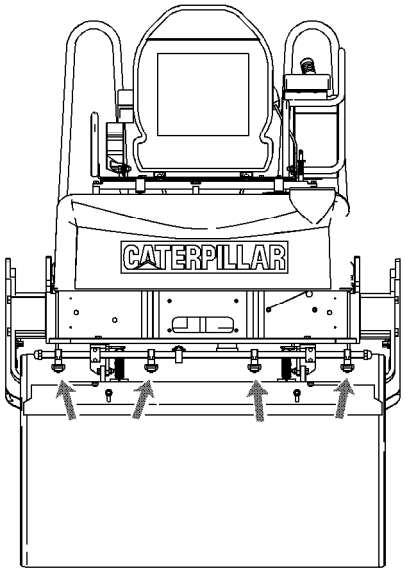


Illustration 128 g01035287

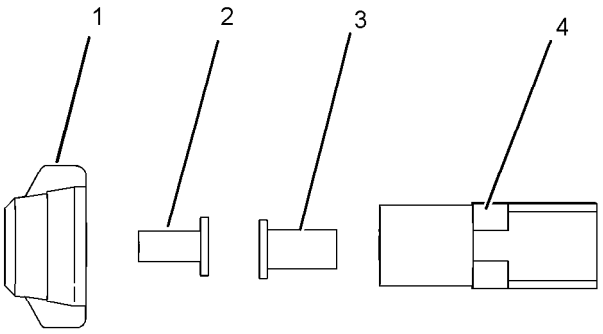


Illustration 129 g01035268

1. Remove cap (1).

2. Remove spray nozzle (2).

3. Remove screen (3).

4. Wash nozzle (2) and screen (3) in a clean, nonflammable solvent.

5. Install screen (3).

6. Install nozzle (2).

7. Install cap (1).

Note: Rotation of the nozzle may be required in order to establish a correct spray pattern.

i01978128

Water Spray System - Drain

SMCS Code: 5612-543

The water system must be drained prior to frost or freezing conditions.

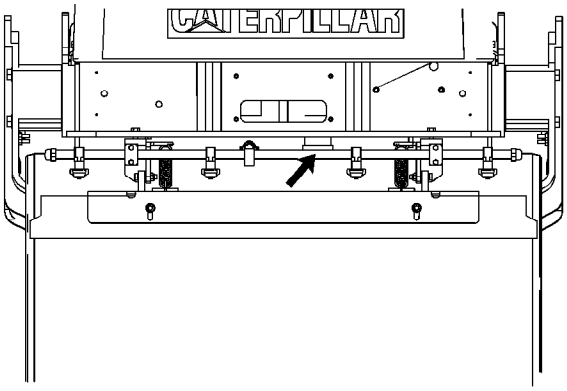


Illustration 130 g01037548

1. Remove the drain plug in order to drain the main water tank.

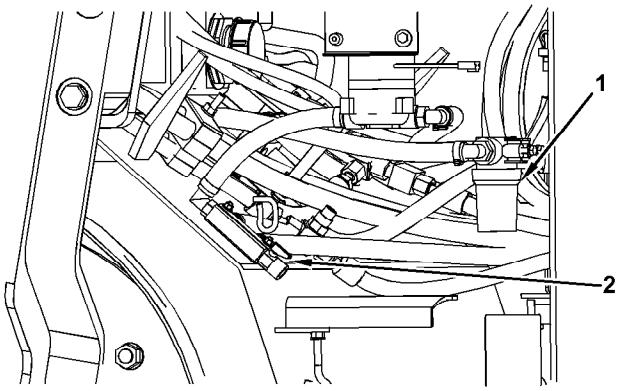


Illustration 131 g01037609

(1) Water filter
(2) Drain cock

2. Open drain cock (2) in order to drain the lines for the spray bar.

3. Access water filter (1) through the step on the right side of the machine.

4. Remove the housing for the water filter.

5. Remove the strainer from the housing.

6. Drain the water from the housing.

7. Allow the water in the water line to drain.

8. Clean the strainer.

9. Install the strainer into the housing for the water filter.

10. Install water filter (1).

Auxiliary Tank (If Equipped)

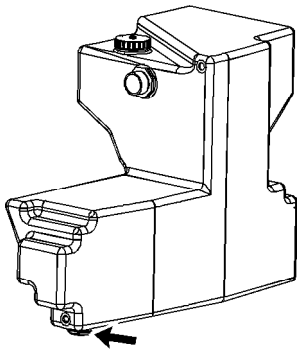


Illustration 132 g01041722

If your machine is equipped with an auxiliary tank, the drain plug is located on the bottom of the tank.

i01978131

Water Spray System Filter - Clean

SMCS Code: 5612-070-FI

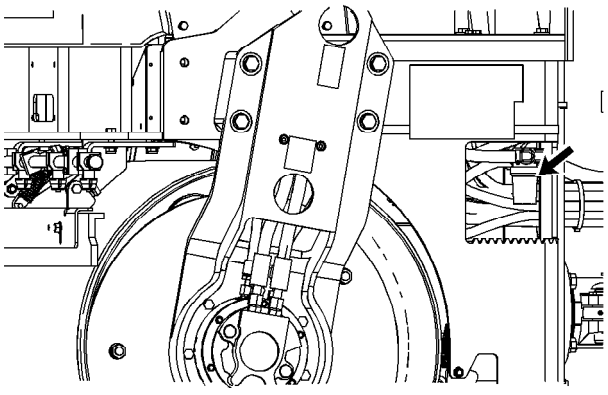


Illustration 133 g01037554

This is the location of the water filter.

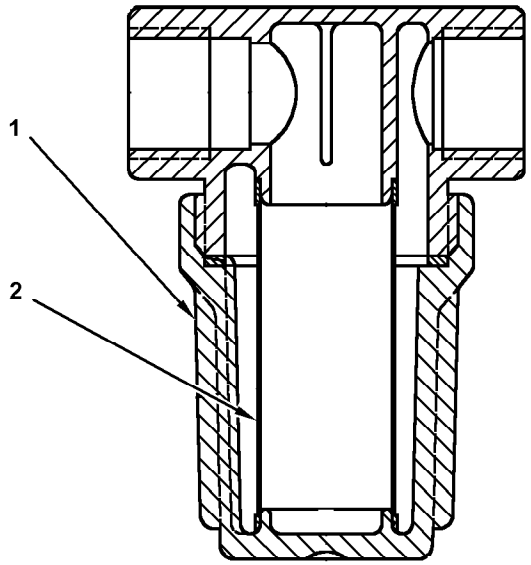


Illustration 134 g01037635

Water Filter

1. Remove filter bowl (1). Remove screen (2).

2. Clean filter bowl (1) with water or compressed air.

3. Clean screen (2) with water or compressed air.

4. Install filter (2) into filter bowl (1).

5. Install filter bowl (3).

i01978132

Water Tank Strainer - Clean and Inspect

SMCS Code: 5613-571-STR

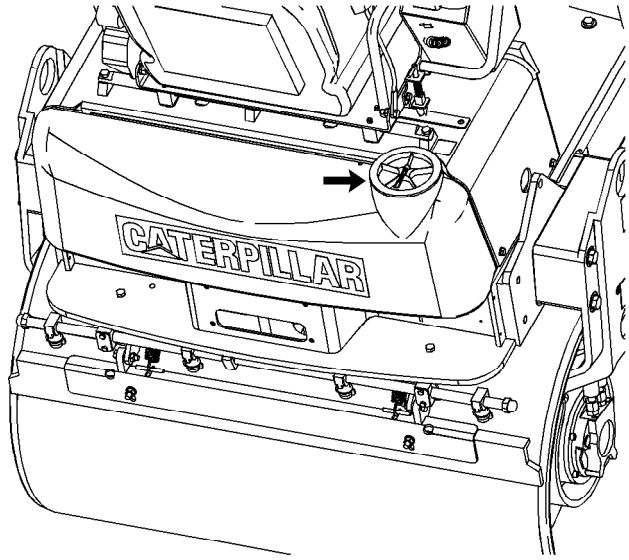


Illustration 135 g01037682

1. Remove the filler cap.

2. Remove the strainer.

3. Clean the filler cap with clean water or compressed air.

4. Clean the strainer with clean water or compressed air.

5. Install the strainer.

6. Fill the tank with clean water.

7. Install the filler cap.

i01978133

Wheel Nuts - Tighten

SMCS Code: 4201-527-NT; 4210-527

S/N: 2251–Up



Illustration 136 g00665858

Tighten wheel nuts to 165 ± 15 N·m (121.7 ± 11 lb ft).

Reference Information Section

Reference Materials

i03044924

Reference Material

SMCS Code: 1000; 7000

The following literature can be obtained from any Caterpillar dealer:

Special Publication, SEBD0518, “Know Your Cooling System”

Special Publication, SEBD0970, “Coolant and Your Engine”

Special Publication, SEBD0717, “Diesel Fuels and Your Engine”

Special Instruction, SEHS9031, “Storage Procedure for Caterpillar Products”

Special Publication, SEBU5898, “Cold Weather Recommendations”

Special Publication, SEBD0640, “Oil and Your Engine”

Special Publication, SEBU6250, “Caterpillar Machine Fluids Recommendations”

Special Publication, SEBU6981, “Emissions Control Warranty Information”

Special Publication, SEBF8029, “Guideline For Reusable Parts and Salvage Operations”

Operation and Maintenance Manual, SEBU8257, “The European Union Physical Agents (Vibration) Directive “2002/44/EC” ”

Service Manual, SENR5664, “Air Conditioning and Heating with R-134A (All Caterpillar Machines)”

Special Instruction, REHS0354, “Charging System Troubleshooting”

Special Instruction, REHS1642, “Operation of the Product Link System”

Special Publication, NENG2500, “Caterpillar Dealer Service Tool Catalog”

Service Magazine, SEBD1587, “What ROPS/FOPS Certification Means”

Service Magazine, SEHS6929, “Inspection, Maintenance and Repair of ROPS and Attachment Installation Guidelines”

Specifications, SENR3130, “Torque Specifications”

Operation and Maintenance Manuals are available in other languages. Consult your Caterpillar dealer for information about obtaining these Operation and Maintenance Manuals.

Additional Reference Material

ASTM D2896, “TBN Measurements” This can normally be obtained from your local technological society, from your local library, or from your local college.

SAE J313, “Diesel Fuels” This can be found in the SAE handbook. Also, this publication can be obtained from your local technological society, from your local library, or from your local college.

SAE J754, “Nomenclature” This can normally be found in the SAE handbook.

SAE J183, “Classification” This can normally be found in the SAE handbook.

Engine Manufacturers Association, “ Engine Fluids Data Book”

Engine Manufacturers Association
Two North LaSalle Street, Suite 2200
Chicago, Illinois, USA 60602
E-mail: ema@enginemanufacturers.org
(312) 827-8700
Facsimile: (312) 827-8737

i03989612

Decommissioning and Disposal

SMCS Code: 1000; 7000

When the product is removed from service, local regulations for the product decommissioning will vary. Disposal of the product will vary with local regulations. Consult the nearest Cat dealer for additional information.

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